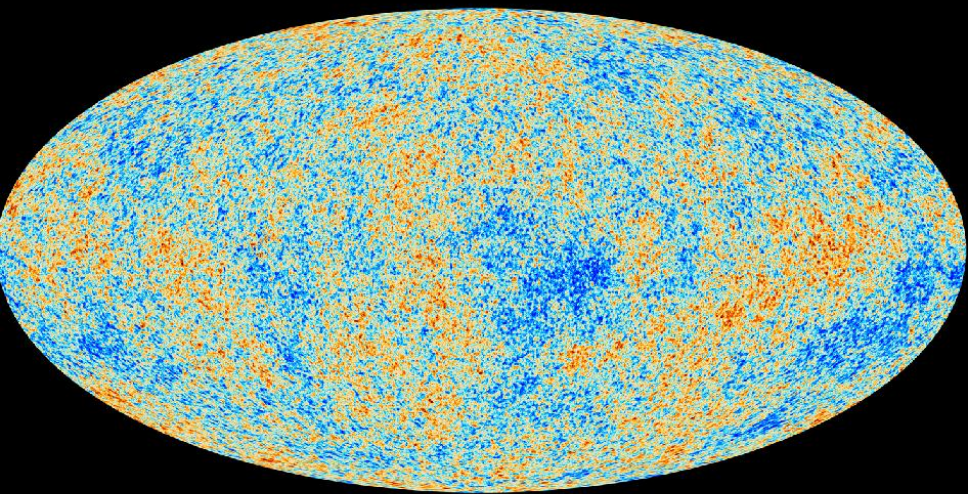




The Big Bang and Inflation

[illegible]

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Astrophysics Department
Cavendish Laboratory
University of Cambridge

16th September 2014

Overview

What is Cosmology?

Our Place in the Cosmos

The Big Bang

The Birth of the Universe

Inflation

What is Cosmology?

Cosmology is the study of the universe as a whole

Age: How old is the universe?

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Content: What are Dark Matter and Dark Energy?

The Night Sky

The Milky Way



The Night Sky

The Milky Way



Ground based telescopes

The Coma Cluster



Ground based telescopes

The Coma Cluster



Space Telescopes

Hubble Extreme Deep Field



Space Telescopes

Hubble Extreme Deep Field



The Dark Ages

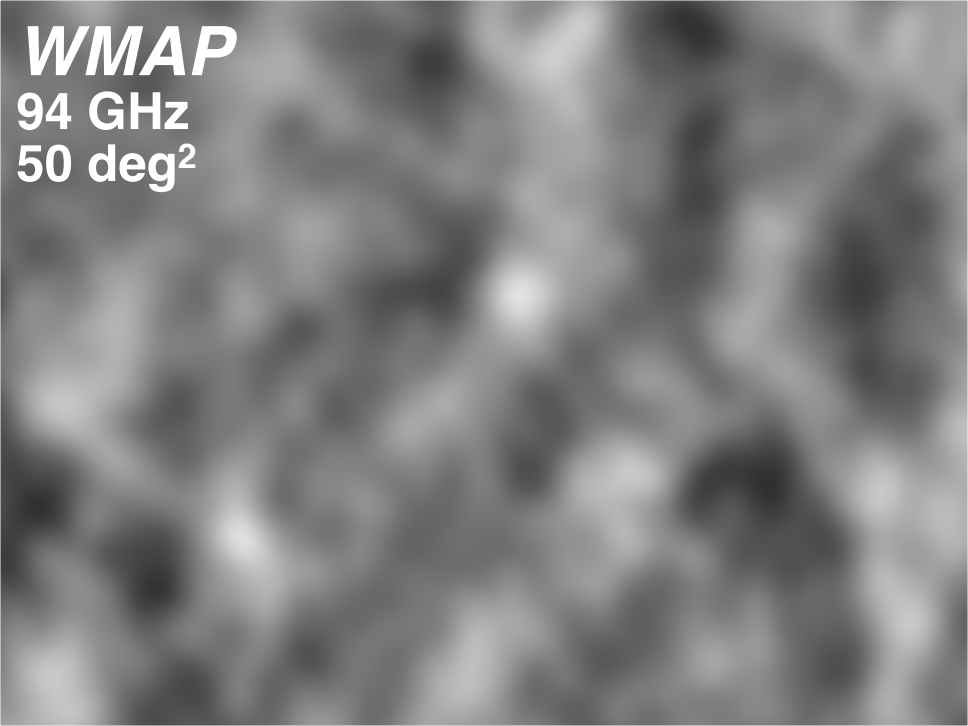
The CMB

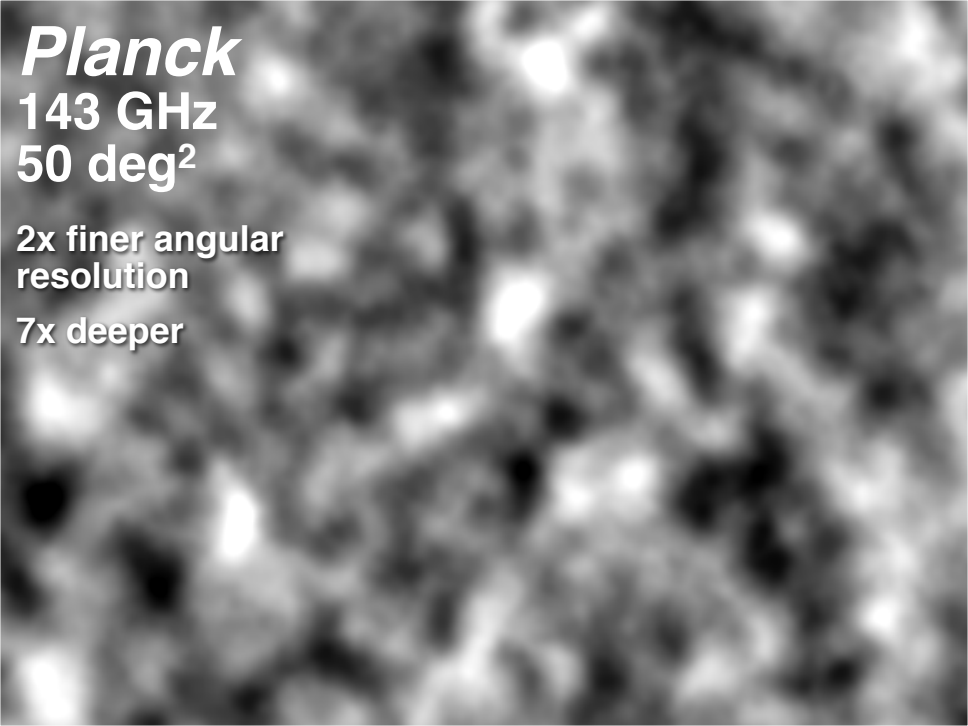
Cosmic Microwave Background

WMAP

94 GHz

50 deg²





Planck

143 GHz

50 deg²

**2x finer angular
resolution**

7x deeper

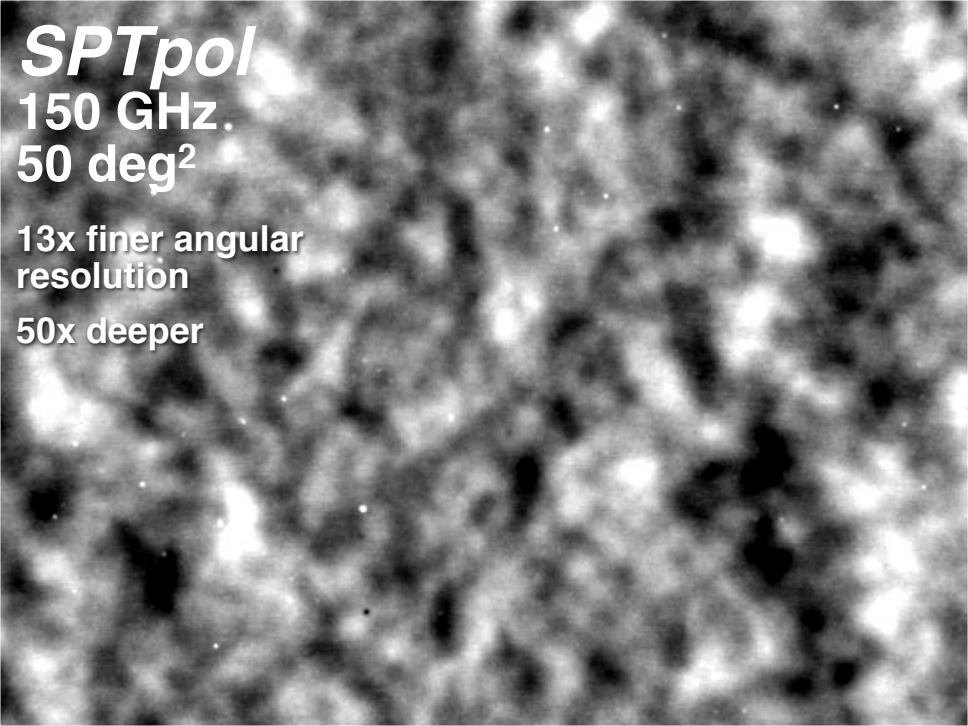
SPTpol

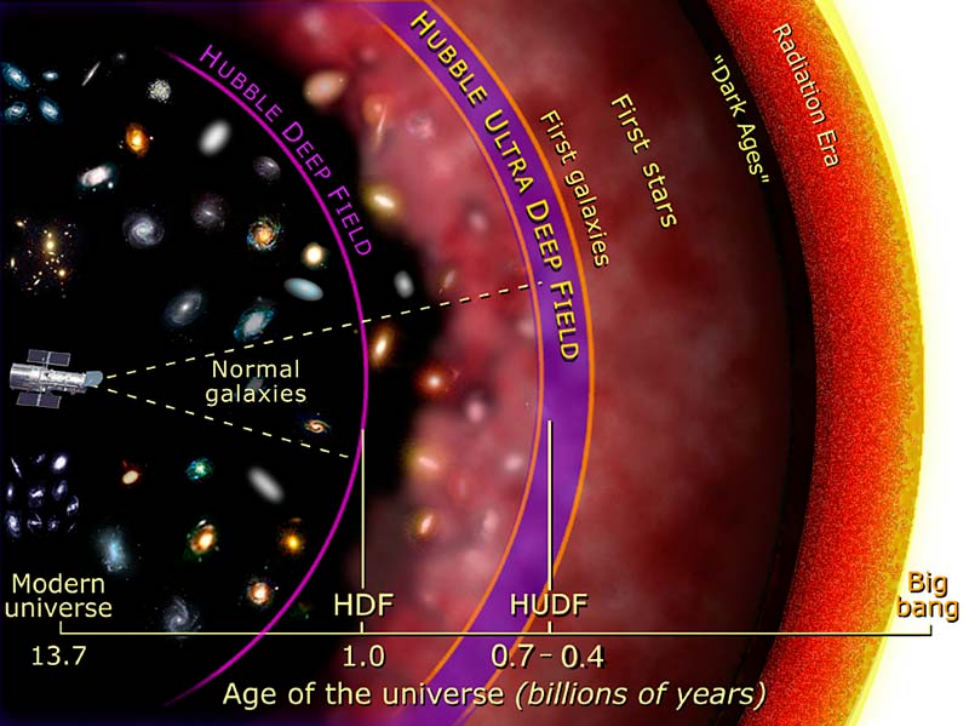
150 GHz

50 deg²

**13x finer angular
resolution**

50x deeper





What is the CMB?

- ▶ In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with “The Big Bang” .

The Big Bang

Galactic Redshift



The Big Bang

Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

The Big Bang

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1. All Galaxies are moving away from us

The Big Bang

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1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away

The Big Bang

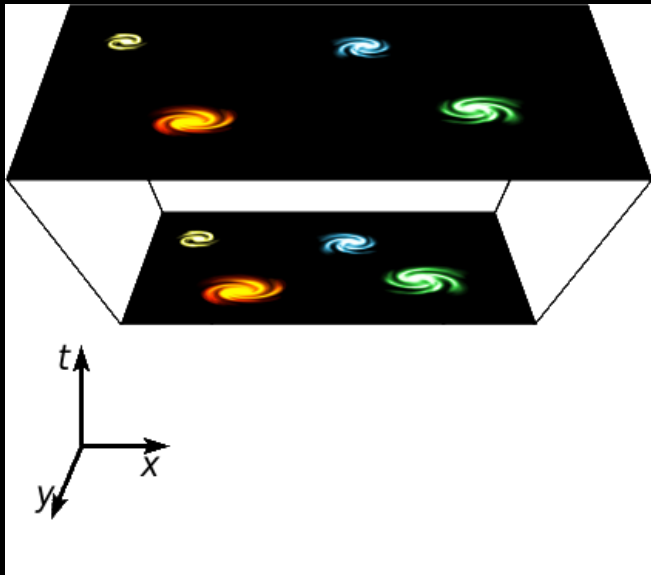
Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

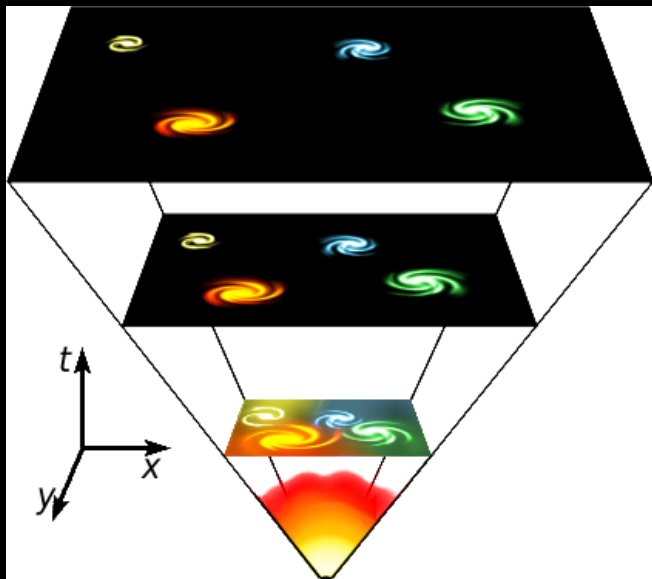
The Big Bang

A stretching universe:



The Big Bang

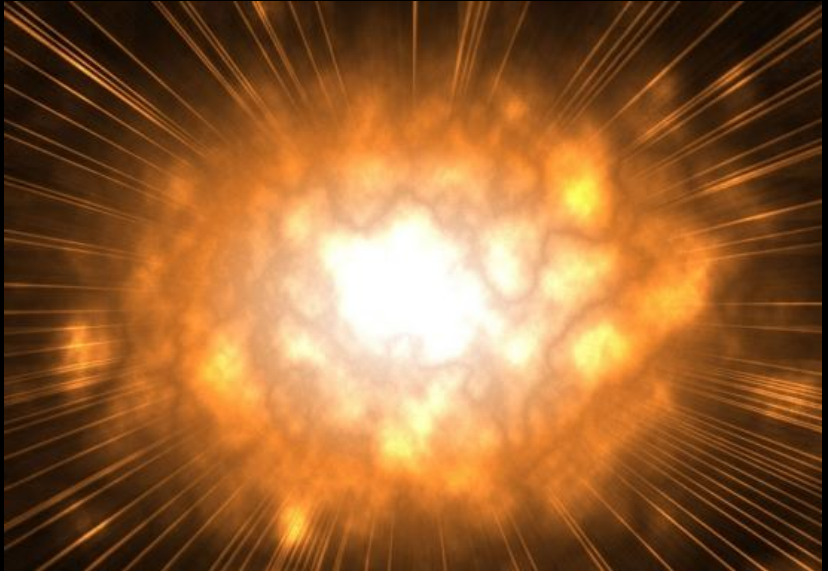
A stretching universe:



The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:



The Big Bang

Common Misconceptions

- ▶ Despite the name, the Big Bang is not an explosion.

The Big Bang

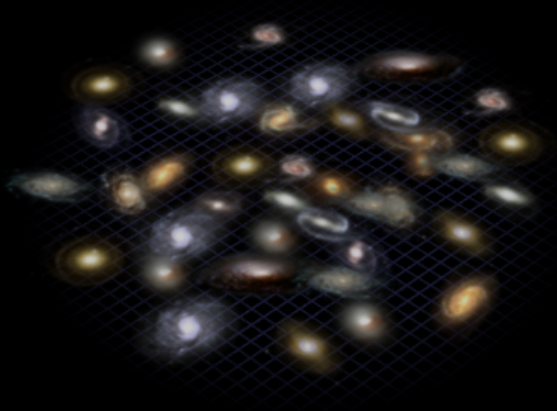
Common Misconceptions

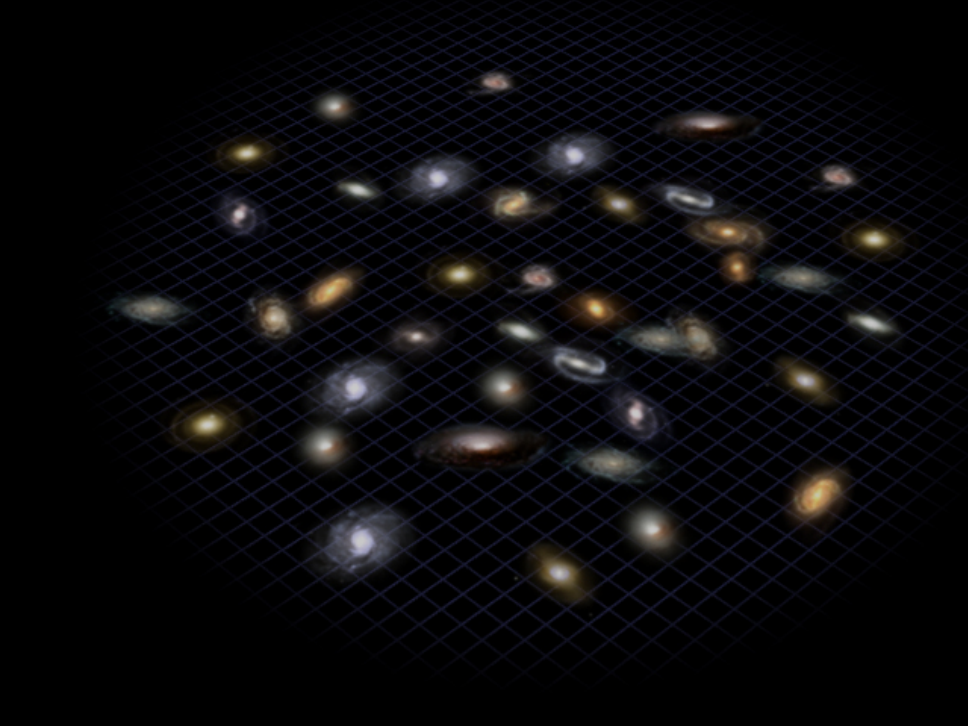
- ▶ Despite the name, the Big Bang is not an explosion.
- ▶ The Universe is not necessarily expanding into anything.

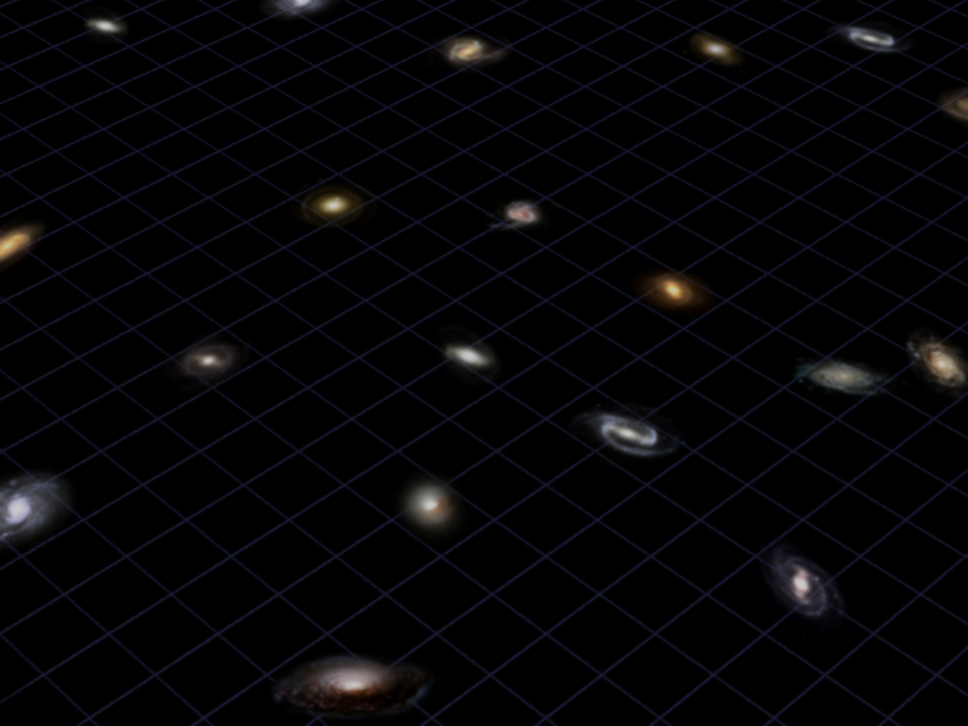
The Big Bang

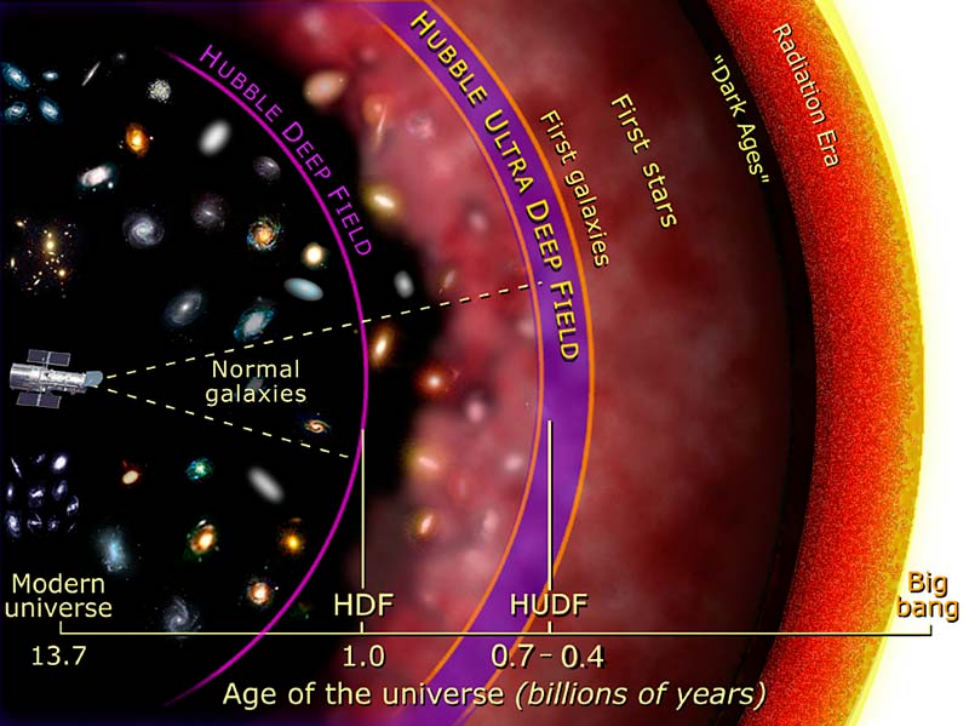
Common Misconceptions

- ▶ Despite the name, the Big Bang is not an explosion.
- ▶ The Universe is not necessarily expanding into anything.
- ▶ The big bang didn't begin at a particular point









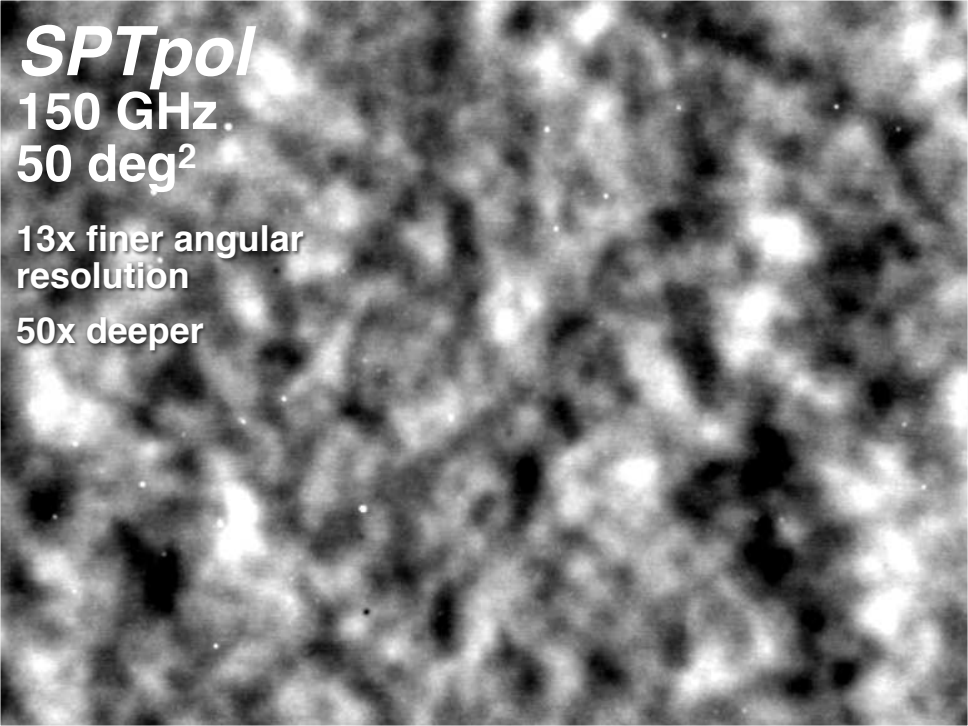
SPTpol

150 GHz

50 deg²

**13x finer angular
resolution**

50x deeper



Current Issues in Cosmology

There are several outstanding problems in Astrophysics which we do not fully understand:

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- ▶ Dark Matter

Current Issues in Cosmology

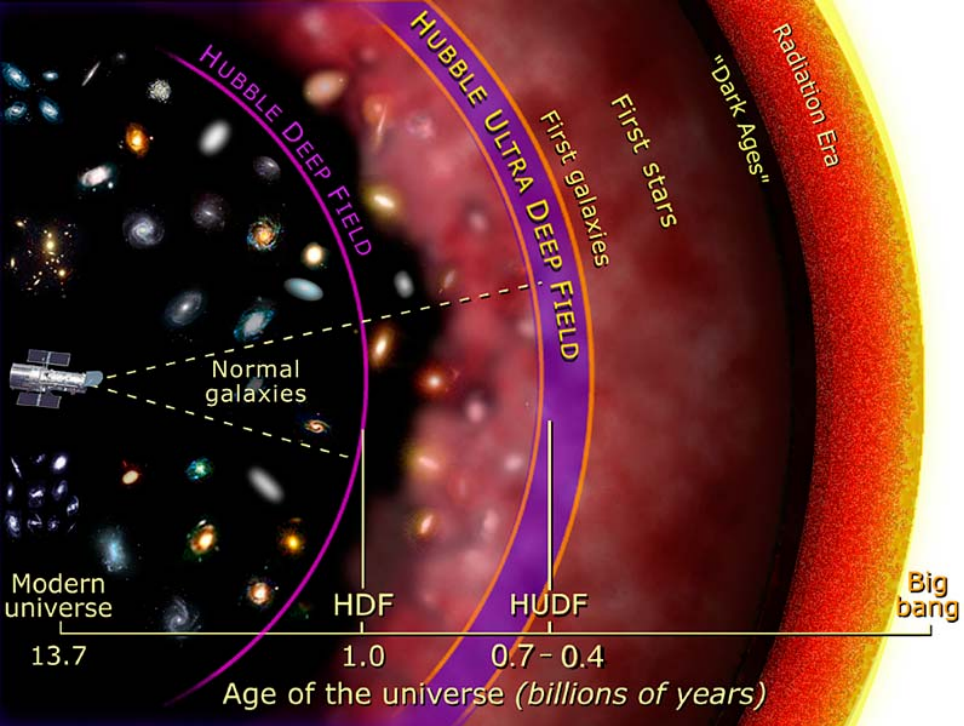
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- ▶ Dark Matter
- ▶ Dark Energy

Current Issues in Cosmology

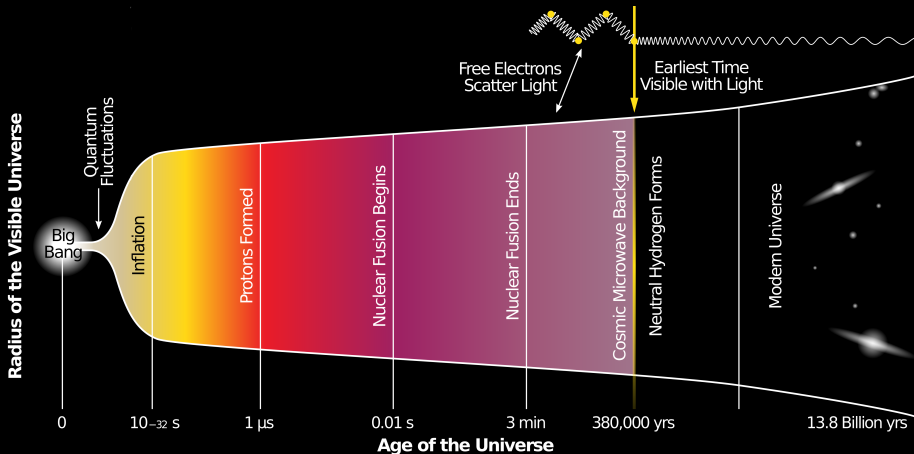
There are several outstanding problems in Astrophysics which we do not fully understand:

- ▶ Dark Matter
- ▶ Dark Energy
- ▶ $t = 0$



The beginning of time

$t = 0$



Problems with basic cosmologies

Problems with basic cosmologies

1. The universe is too smooth

Problems with basic cosmologies

1. The universe is too smooth (homogeneous)

Problems with basic cosmologies

1. The universe is too smooth (homogeneous)
2. The universe is too flat

The universe is too smooth (homogeneous)

- ▶ The CMB is far too uniform

The universe is too smooth (homogeneous)

- The CMB is far too uniform

The CMB

Cosmic Microwave Background

SPTpol

150 GHz.

50 deg²

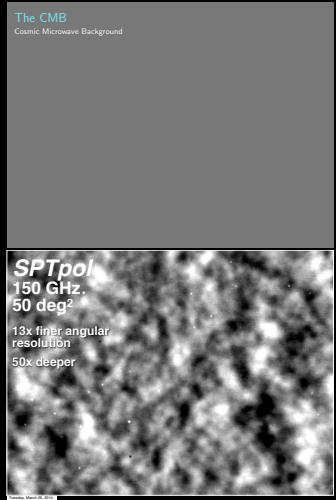
13x finer angular
resolution

50x deeper

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The universe is too smooth (homogeneous)

- ▶ The CMB is far too uniform
- ▶ In order for it to be that smooth, the universe would have to be much older.

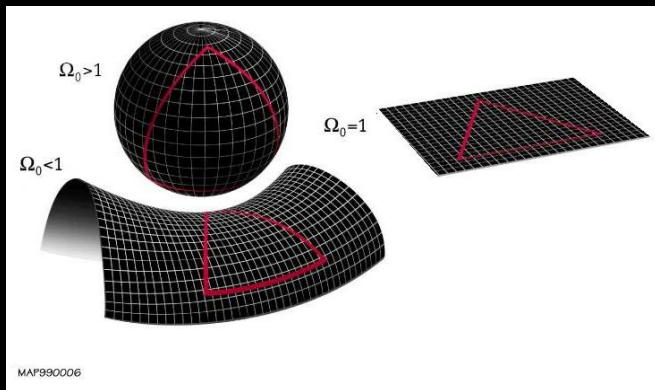


The universe is too flat

- ▶ The universe can come in three shapes

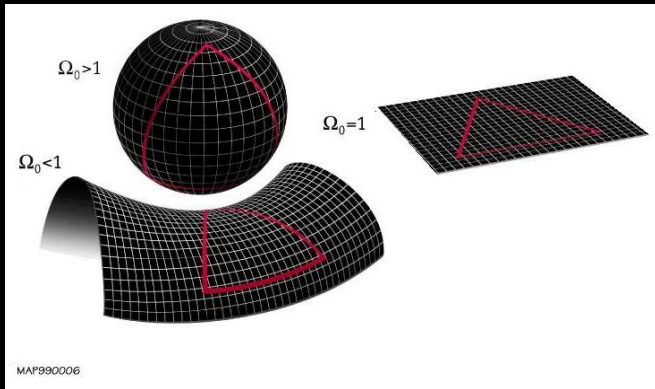
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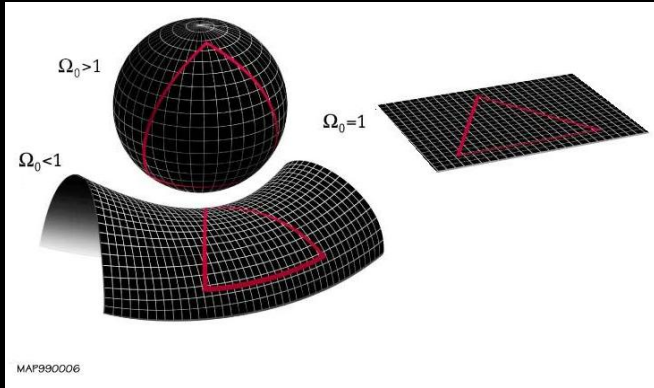
- The universe can come in three shapes



1. Spherical
(closed)

The universe is too flat

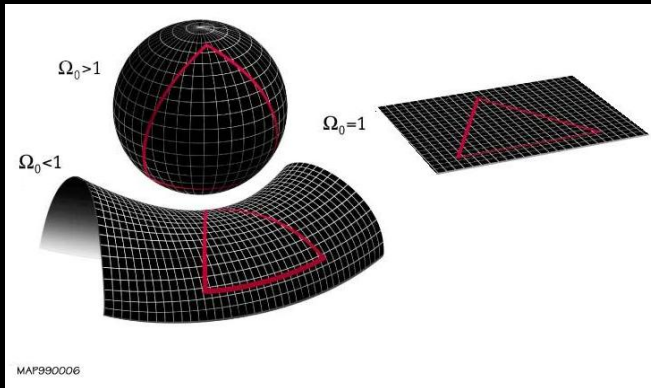
- The universe can come in three shapes



1. Spherical
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2. Flat

The universe is too flat

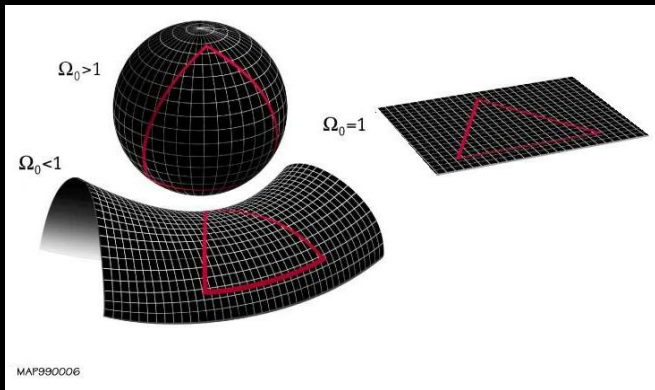
- ▶ The universe can come in three shapes



1. Spherical (closed)
2. Flat
3. Saddle (open)

The universe is too flat

- ▶ The universe can come in three shapes



1. Spherical
(closed)

2. Flat

3. Saddle (open)

- ▶ Flat is exceedingly unlikely

Inflation

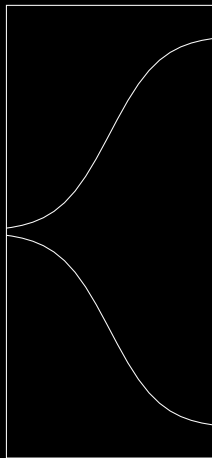
A solution

- ▶ Early on in history, the universe expanded incredibly rapidly

Inflation

A solution

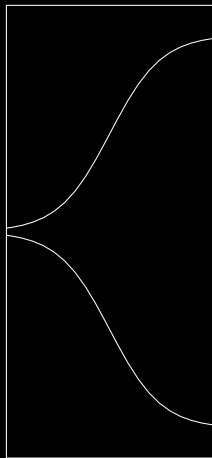
- ▶ Early on in history, the universe expanded incredibly rapidly



Inflation

A solution

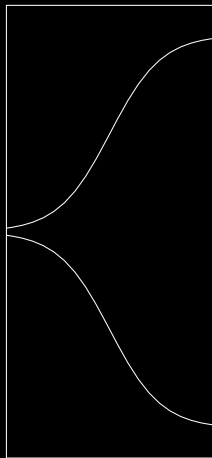
- ▶ Early on in history, the universe expanded incredibly rapidly
- ▶ We call this accelerated expansion “Inflation”



Inflation

A solution

- ▶ Early on in history, the universe expanded incredibly rapidly
- ▶ We call this accelerated expansion “Inflation”
- ▶ Virus $\rightarrow$ Galaxy in 10^{-32} seconds



Evidence for inflation

The CMB

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© 2013, March 29, 2013

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- Not only does inflation explain why the CMB is so smooth. . .

Evidence for inflation

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© 2013, South Pole, 2013

- ▶ Not only does inflation explain why the CMB is so smooth...
- ▶ ... It also explains the small fluctuations.

Big Bang

Big Bang plus
 10^{-43} seconds

quantum-gravity era

inflation

Inflation acts as a 'cosmic microscope'

Big Bang plus
 10^{-35} seconds?

cosmic microwave background

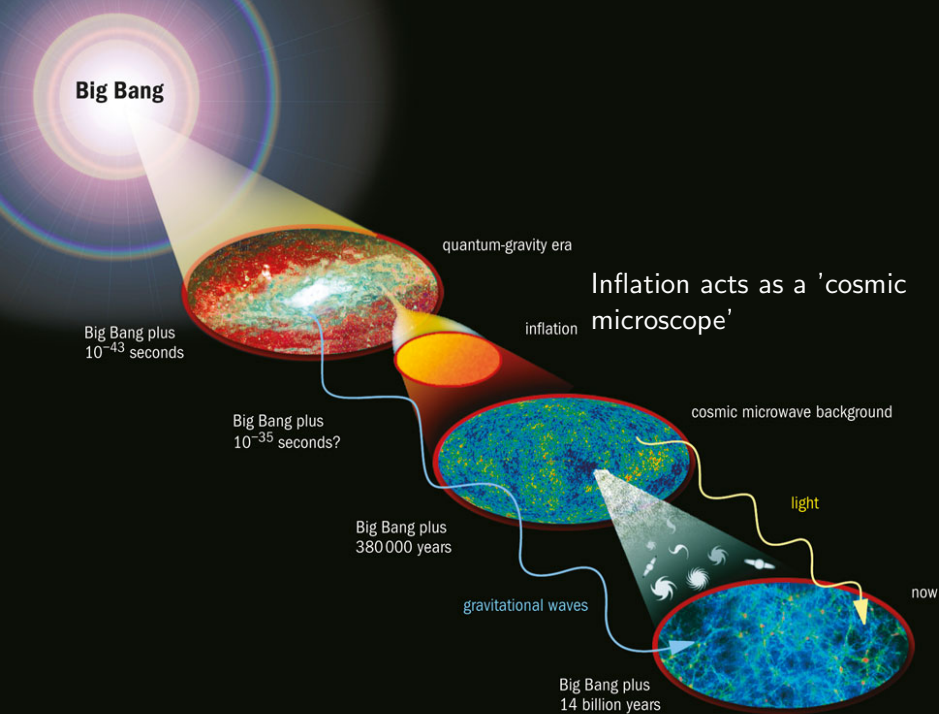
Big Bang plus
380 000 years

light

gravitational waves

now

Big Bang plus
14 billion years



Big Bang

Big Bang plus
 10^{-43} seconds

quantum-gravity era

inflation

It magnifies quantum
patterns to cosmic scales

Big Bang plus
 10^{-35} seconds?

cosmic microwave background

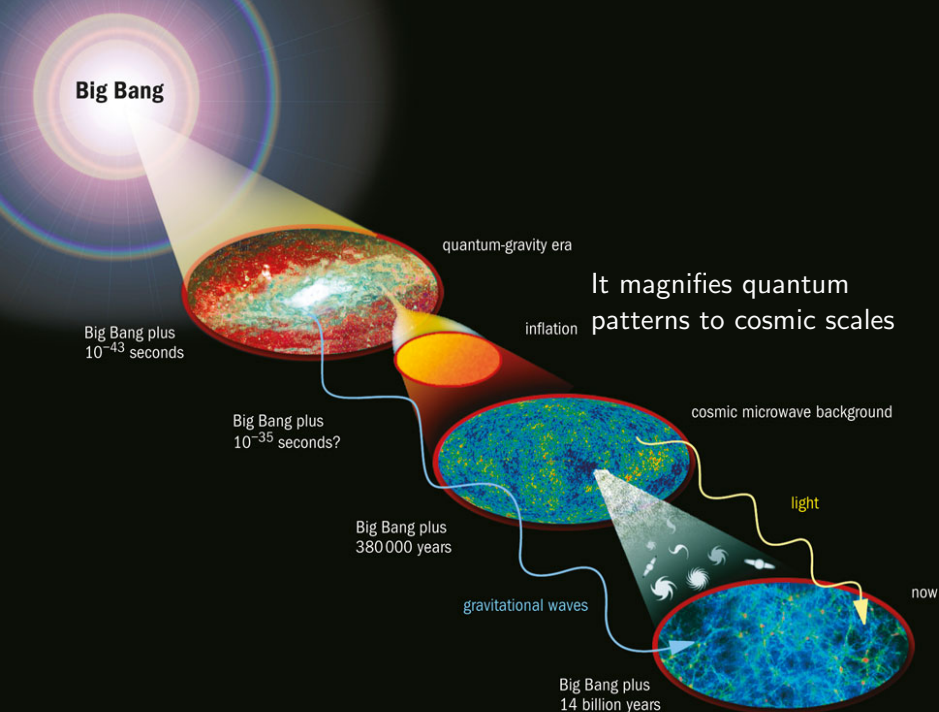
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The physics of the
smallest scales is splashed
across the stars

Big Bang plus
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cosmic microwave background

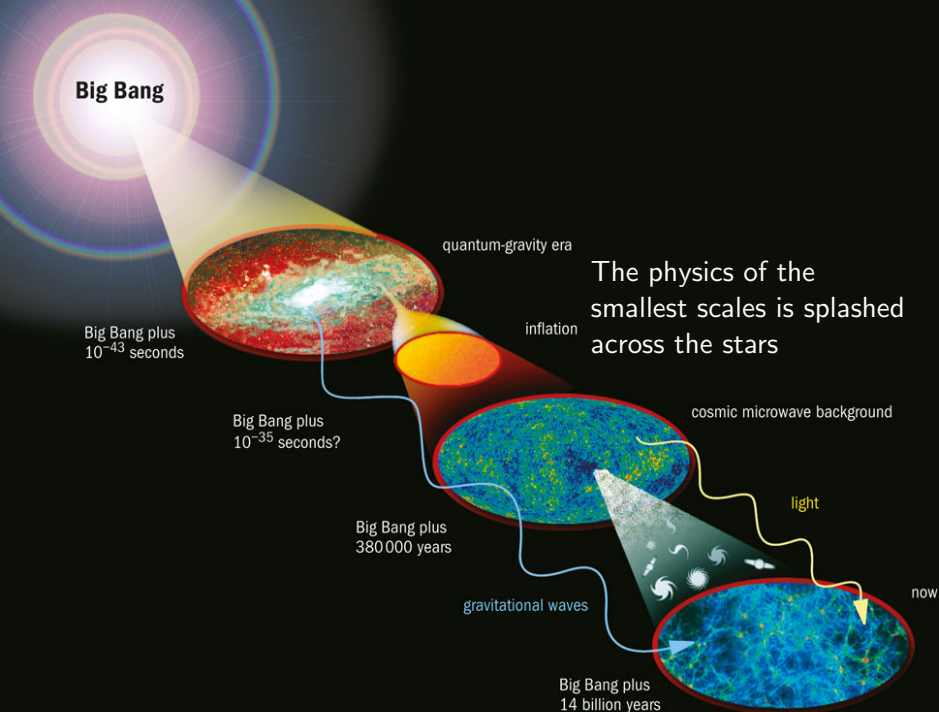
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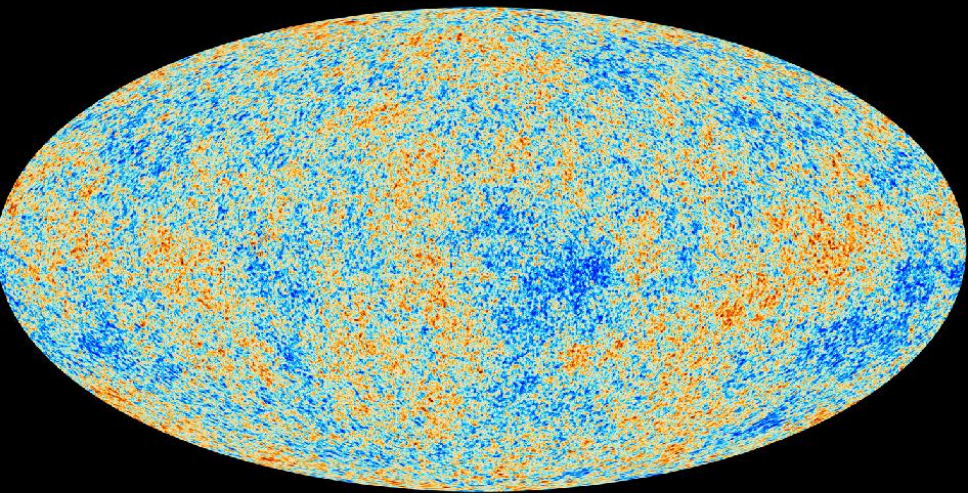
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Thank you!

Questions?



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